

Problem 2.11

Suppose that you invest \$4,000 at time 0 into an investment account with an accumulation function of

$$a(t) = \alpha t^2 + 4\beta.$$

At time 4, your investment has accumulated to \$5,000. Find the accumulated value of your investment at time 10.

Solution

We are given the accumulation function

$$a(t) = \alpha t^2 + 4\beta.$$

Since an accumulation function must satisfy

$$a(0) = 1,$$

we have

$$a(0) = \alpha(0)^2 + 4\beta = 4\beta = 1.$$

Therefore,

$$\beta = \frac{1}{4}.$$

So the accumulation function becomes

$$a(t) = \alpha t^2 + 1.$$

Now use the information that \$4,000 invested at time 0 accumulates to \$5,000 at time 4:

$$A(0) a(4) = A(4).$$

Thus,

$$4000(16\alpha + 1) = 5000.$$

Divide both sides by 4000:

$$16\alpha + 1 = \frac{5000}{4000} = \frac{5}{4}.$$

Hence,

$$16\alpha = \frac{5}{4} - 1 = \frac{1}{4},$$

so

$$\alpha = \frac{1}{64} = 0.015625.$$

Therefore,

$$a(t) = 0.015625 t^2 + 1.$$

Now evaluate at $t = 10$:

$$a(10) = 0.015625(10)^2 + 1 = 0.015625(100) + 1 = 1.5625 + 1 = 2.5625.$$

So the accumulated value at time 10 is

$$A(10) = 4000 \cdot a(10) = 4000(2.5625) = 10250.$$

Hence, the accumulated value at time 10 is

$$\boxed{\$10,250}.$$